Sravan
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Data Science/ML Engineer

PROFESSIONAL SUMMARY

Experienced **Data Science and MLOps** professional with 4+ years of experience with a strong mathematics, statistics, and programming background. Proficient in extracting insights from complex datasets and developing predictive models to drive data-driven decision-making. Skilled in data engineering, data cleaning, preprocessing, exploratory data analysis, designing and managing scalable infrastructure, deploying, and serving models, building robust data pipelines, implementing CI/CD processes, and applying statistical techniques and machine learning algorithms. Expertise in data visualization and effectively communicating findings to stakeholders. Collaborative team player with excellent problem-solving skills and a passion for continuous learning and research. Strong ability to translate business requirements into actionable analytical solutions.

- Skilled in applying statistical techniques and machine learning algorithms to develop models that can predict outcomes, classify data, segment populations, and make data-driven decisions.
- Experience applying a wide range of machine learning algorithms and developing various **Regression, Classification, Time series, Anomaly detection, Propensity, Market Mix Models, Multi-touch attribution, NLP, Deep Learning, and LLM models.**
- Developed expertise in emergent **Generative AI** technologies including **large language models** using transformer mechanisms (BERT, GPT, T5). Leveraged **LangChain**, an open-source conversational framework, to rapidly prototype and evaluate custom conversational agents for enhancing business productivity.
- Experience in designing and building **Machine Learning pipelines (MLOps**) in Cloud environments like **AWS (AWS Sagemaker, AWS EC2, AWS EMR, AWS Lambda, AWS Glue)**, **Azure (DevOps, Data Bricks, Data Factory, MLFlow) and GCP (BigQuery)**
- Expertise in **Natural Language Processing (NLP)** in **python** including tokenization, lemmatization, stemming, parts of speech tagging, sentimental analysis.
- Proficient in creating visualization reports and dashboards using Business intelligence tools like **Power BI, Tableau & and Python visualization libraries** like **Seaborn, matplotlib,** and **Bokeh** to effectively communicate the data findings to stakeholders enabling better understanding and decision-making.
- Engineered and managed the deployment infrastructure for machine learning and LLMs using **Docker** and **Kubernetes**, enhancing the operational efficiency of ML & AI-powered services.

TECHNICAL SKILLS

Project Management tools	Microsoft Project, Jira, Confluence, Agile, Scrum, SharePoint
Database	MS SQL, Big Query, Postgres SQL
Programming	SQL, T-SQL, Python (Pandas, NumPy), R, PySpark, PySQL, SnowSQL, Scala
Versioning Tools	Git, Docker, DVC
MLOps tools	MLFlow, AutoML, Azure MLOps, Azure ML Designer
Machine Learning Algorithms	Classification, Regression, Forecasting, Decision Trees, K-Means, SVM, Boosting and Bagging methods, NLP (Sentiment Analysis, Semantic search), Market Mix Modelling (MMM), Multi-touch attribution models, Generative AI, and Deep Learning. LLM (BERT/GPT/T5), Prompt engineering
Machine Learning Libraries	NumPy, Pandas, PyCaret, Scikit-learn, TensorFlow, PyTorch, Keras, Transformers, SpaCy, Matplotlib, Seaborn

Statistical Methods	A/B Testing, T-test, Chi-square test Predictive Analysis, Hypothesis Testing, PCA, MBA, Text Analytics, ANOVA, Text Mining (Bag of Words, TF-IDF)
Data Visualization	Power-BI, Tableau, Google Looker Studio
Cloud Computing	Azure (DevOps, Data Lake Storage, ML Studio, Data Factory, Data Bricks, Synapse), Amazon Web Services (EC2, S3, Glue, Lambda, Sage Maker, EMR), GCP (ML, BigQuery), Snowflake

PROFESSIONAL EXPERIENCE

Client: CVS Health, Chicago, IL **Oct 2023 - Present**
Role: Consultant – Data Science & ML
Responsibilities:

- Leveraged deep domain knowledge and expertise in **data science, MLOps, and data engineering** on the **Azure Cloud** platform to provide strategic guidance and insights on complex data-driven projects.
- Collaborated with cross-functional teams to develop a **GenAI** HR application for Job profile creations leveraging OpenAI models such as **GPT-4** and **LangChain** as a framework to generate highly contextual and domain-specific content.
- Developed and implemented a propensity model framework using **XGBoost** on **Azure Databricks** to predict the likelihood of conversion enabling business to prioritize leads that increased sales revenue by 26% through accurate forecasting of market trends.
- Worked on building customer segmentation models using **K-means clustering** and **DBSCAN** machine learning techniques utilizing Adobe Analytics through Python helping businesses to identify the target audience and tailor personalization campaigns.
- Developed timeseries forecasting models in **python** using stats models and **TensorFlow**
- Worked on **Amazon SageMaker,** a fully managed service that helped us prepare, build, train, and deploy high-quality machine learning models quickly by bringing together a broad set of capabilities.
- Worked on the development and deployment of a demand **forecasting model** using **deep learning** techniques, Employed **LSTM neural networks** within TensorFlow and Keras to model and predict future product demands based on historical sales data,
- Performed data preprocessing, feature engineering, and data transformation tasks on **Azure Databricks** to prepare raw data for model training using **Spark SQL** and **PySpark** by collaborating with business stakeholders mainly from sales and marketing domains to provide data-driven solutions.
- Designed and implemented a comprehensive data integration pipeline using **Azure Data Factory** to seamlessly consolidate data from multiple sources to orchestrate the data flow into **Azure Synapse Analytics**.
- Employed **Docker** for containerization and **Kubernetes** for orchestration, integrating Helm charts and Docker Compose for efficient service deployment and management.
- Deployed machine learning models as **RESTful web services** using Azure services like **Azure ML Studio** and worked on building **CI/CD pipelines** for implementing monitoring solutions to track model drift, retrain models when necessary, and ensure continued model performance using **Azure DevOps** service.
- Worked on building **MLOps frameworks** using **MLFlow** facilitating the seamless integration of machine learning into the production environment, enabling rapid model deployment, real-time monitoring for **data drift and model drift**, and cost-effective operations.
- Used AWS **SageMaker** for developing end to end ML pipelines.
- Used **AWS Lambda** and **Glue** for creating highly functional Data pipelines.
- Developed BI dashboards using **PowerBI** server showing the business an overall view of KPIs to understand customer behavior and generate insights related to various metrics like Unique visitors, Bounce rate, demographics, and conversions improving the customer journey and ROI.

- Analyzed the marketing campaigns (Campaign Management) on a weekly/monthly basis by pulling data from **Azure SQL DB** using **Azure Databricks** to give business insights on various KPIs (Conversion rate, ROI, Customer acquisition, Channel Performance) by gathering information from multiple data sources (web analytics data as well as external data)
- Developed scripts for build, deployment, maintenance, and related tasks using Jenkins, Docker, Python and Bash.

Environment: Python, MLFlow, SQL, Spark, Excel, Machine Learning libraries (Scikit-learn, PyTorch, Keras & TensorFlow), LLM (GPT-4), NLP, Forecasting, LangChain, AWS SageMaker, PowerBI, AWS Lambda, Azure (Databricks, ML Studio, Data Factory, Azure DLS Gen2, Devops), Generative AI, Docker, Kubernetes

Client: Xempler Insights, India **Mar 2021 - Jun 2023**
Role: Data Scientist
Responsibilities:

- Partnered with the Marketing Analytics team from AdTech and retail clients in the US to build a **Market Mix Model (MMM)** and **multi-touch attribution model** using **Lightweight MMM** using **Azure ML Studio** to quantitatively estimate the effectiveness of various marketing elements, identify key sales drivers and the high ROI Channels.
- Utilized **Apache Hive** and **Spark** to manage and analyze large datasets in a predictive maintenance machine learning with a focus on developing and optimizing ETL processes in Hive to structure and query big data efficiently, while employing Spark's MLlib in Databricks for building and training predictive models.
- Developed and managed a comprehensive data pipeline in **Azure Databricks**, integrating diverse source systems to populate a **feature store**, enabling efficient model development and deployment.
- Worked on deployment and optimization of a **Convolutional Neural Network (CNN)** model using **TensorFlow** for product image classification task, leveraging **Azure cloud platform** capabilities. This initiative streamlined the automation of product categorization, significantly improving operational efficiency and accuracy in inventory management.
- Built a customer attrition Random Forest Model with **AWS Sage maker** that improved the monthly retention by 12 basis points for client likely to opt-out by providing relevant product features for them.
- Performed research for an AIOps POC project on the development of a cutting-edge **large language model (LLM)** using **LangChain**, **Azure OpenAI** & **Azure CogSearch** with **RAG** Capability with a primary focus on developing chat-based applications on private & organizational data.
- Worked with **Artificial Neural Networks (ANN)** and Deep Learning models using **Theano**, **TensorFlow** and **Keras** packages using **Python**.
- Performed Data Wrangling, feature engineering, apply statistical analysis, data visualization, data mining using Python (**Pandas, NumPy, Matplotlib, seaborn, Scikit-learn**).
- Designed and implemented a **microservices architecture** for **MLOps** to facilitate the seamless scaling and management of ML models, significantly reducing response latency and improving operational efficiency.
- Implemented **spark-based partitioning approach** for data lakes stored in **Databricks**, optimizing data workloads within the Databricks platform, enhancing query performance, and reducing processing times significantly.
- Worked on the development of a **Large Language Model (LLM)** with a focus on enhancing Natural Language Understanding (NLU). Collaborated with a multidisciplinary team to design and implement architecture tailored for **Sentiment analysis, and semantic search** models using **BERT**.

Environment: Python, MLFlow, SQL, Spark, Excel, Deep Learning (Pytorch/TensorFlow), Vector Databases, NLP, LangChain, AWS Lambda, Azure (Databricks, ML Studio, Data Factory, Azure DLS Gen2, Cog Search), Docker, MMM, Generative AI.